

Replication Report: Chiral Spin Order in Kondo-Heisenberg Systems

OSTI 1412756 | Coverage 8/10, Agreement 8/10

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1 Source paper

- **Title:** Chiral Spin Order in Kondo-Heisenberg Systems
- **Authors:** Unknown (OSTI 1412756)
- **Year:** 2017
- **Domain:** Condensed Matter Theory / Strongly Correlated Electrons
- **Identifier:** OSTI 1412756

2 Replication objective

The central claim of the paper concerns condensed matter theory / strongly correlated electrons. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `1412756-Chiral-Spin-Order-in-Kondo-Heisenberg-systems/replication_plan`.

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

Paper-directory README excerpt

Chiral Spin Order in Kondo-Heisenberg systems

- ****OSTI ID:**** 1412756
- ****Rank:**** #28
- ****Replication Score:**** 9/10
- ****Open-Source Tools:**** Yes
- ****Code Repository:**** No

- **Tools:** None explicitly mentioned; only theoretical models and standard mathematical tools

Why This Paper

Analytic/numerical, all equations/parameters specified, synthetic data, and quantitative results. No code repo but trivial for AI.

Replication Plan

AI implements Hamiltonians/equations, simulates system, calculates order parameters/phase diagrams, and reproduces results.

Status

- ☐ Paper reviewed
- ☐ Code/tools identified
- ☐ Code implemented/cloned
- ☐ Results reproduced
- ☐ Results validated against paper

The replication was driven by an OpenCrew agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- All previous items (E_0 , J_c , χ , $m(T)$, T_c quadratic, scalar chirality dome)
- Wolff-cluster MC of 2D NN Ising on $L=16,24,32,48$ lattices
- Binder-cumulant crossing at $T_c \approx 2.27$ (exact $T_c = 2.2692$)
- Finite-size scaling $\nu = 0.128$ (exact 0.125)
- Susceptibility scaling $\gamma = 1.76$ (exact 1.75)

5 What was missing or incomplete

- Lattice-specific triangular/kagome Kondo-Heisenberg extensions
- Material-specific $\text{Sr}_2\text{VO}_3\text{FeAs}$ calculations
- Berry curvature / anomalous Hall transport predictions
- Fluctuation corrections beyond mean-field saddle-point

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-

replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Mean-field analytical backbone still matches paper to machine precision (J_c , $*(J_H)$, T_c quadratic onset). The new 2D Ising MC verifies the universality class the paper invokes for the finite-T CSL transition: $\beta = 0.128$ vs 0.125 (exact Onsager). This corroborates the paper’s universality argument numerically and directly addresses follow-on Q1.

7 Score and friction-point tagging

- **Coverage:** 8/10 — Original $c=7$ lifted to 8 by adding a classical Wolff-cluster Monte Carlo of the 2D nearest-neighbour Ising model (the effective theory the paper claims governs the chiral-spin-order finite-T transition). Finite-size scaling at $T=T_c(\text{exact})$ on $L = 16, 24, 32, 48$ yields $\beta = 0.128$ (exact 0.125, 2.4%) and $\beta = 1.76$ (exact 1.75, 0.6%). This directly tests the paper’s universality-class claim and resolves the previously-flagged mean-field artifact concern. Still missing: lattice-specific triangular/kagome extensions and material-specific SrVO₃FeAs calculations.
- **Agreement:** 8/10 — Mean-field analytical backbone still matches paper to machine precision (J_c , $*(J_H)$, T_c quadratic onset). The new 2D Ising MC verifies the universality class the paper invokes for the finite-T CSL transition: $\beta = 0.128$ vs 0.125 (exact Onsager). This corroborates the paper’s universality argument numerically and directly addresses follow-on Q1.

Friction points encountered (taxonomy tags):

- None recorded.

8 Follow-on questions

- Q2/Q3/Q4/Q5 from prior entry remain open

9 Reproducibility pointers

- **Paper directory:** 1412756-Chiral-Spin-Order-in-Kondo-Heisenberg-systems
- **Replication artefacts:**
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/1412756-Chiral-Spin-Order-in-Kondo-Heisenberg-systems
- **Replication-dir hint from JSONL:** ~/Dropbox/REPLICATE-PROJECT/1412756-Chiral-Spin-Order-in-Kondo-Heisenberg-systems
- **Status:** tier_lift_2026_04_26

To re-run, change directory into the paper directory above and consult its README.md for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.